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Schutte et al.

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(54) **ELECTRICAL SURGE SUPPRESSOR**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 189 days.

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H02H 1/06 (2006.01)
H02H 9/04 (2006.01)

(52) **U.S. Cl.**
CPC **H02H 9/042** (2013.01); **H02H 1/06** (2013.01)

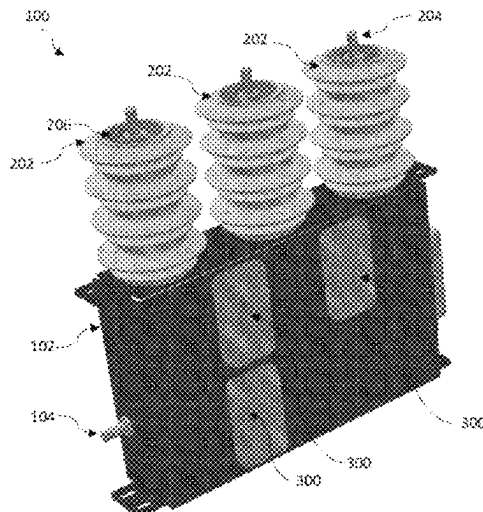
(58) **Field of Classification Search**

CPC H02H 9/04; H02H 1/06; H02H 9/042;
H02H 1/0061; H02H 9/041; H02H 9/005;
(Continued)

(57) **ABSTRACT**

An electrical surge suppressor includes a capacitive component incorporated inside a housing and immersed in, or filled with, oil; a linear resistive component; and a non-linear resistive component connected or connectable in parallel with the linear resistive component. The surge suppressor has a monitoring circuit configured to measure at least one operating characteristic of the linear and/or non-linear resistive and capacitive components and an auxiliary power supply integrated within the capacitive component to provide an auxiliary supply voltage to the monitoring circuit. The capacitive component is connected in series with the parallel connected linear and non-linear resistive components and connected to a phase contact; the phase contact is connected or connectable to the load circuit; and the capacitive component is configured to provide a high impedance connection to the linear and non-linear resistive components, at nominal power supply frequencies and, during switching transients, to provide a low impedance to a combined resistive impedance of the linear and non-linear resistive components effectively into the load circuit.

21 Claims, 13 Drawing Sheets



CPC G01R 31/64; G01R 27/26; G01R 15/06;
G01R 31/12
USPC 361/117-120
See application file for complete search history.

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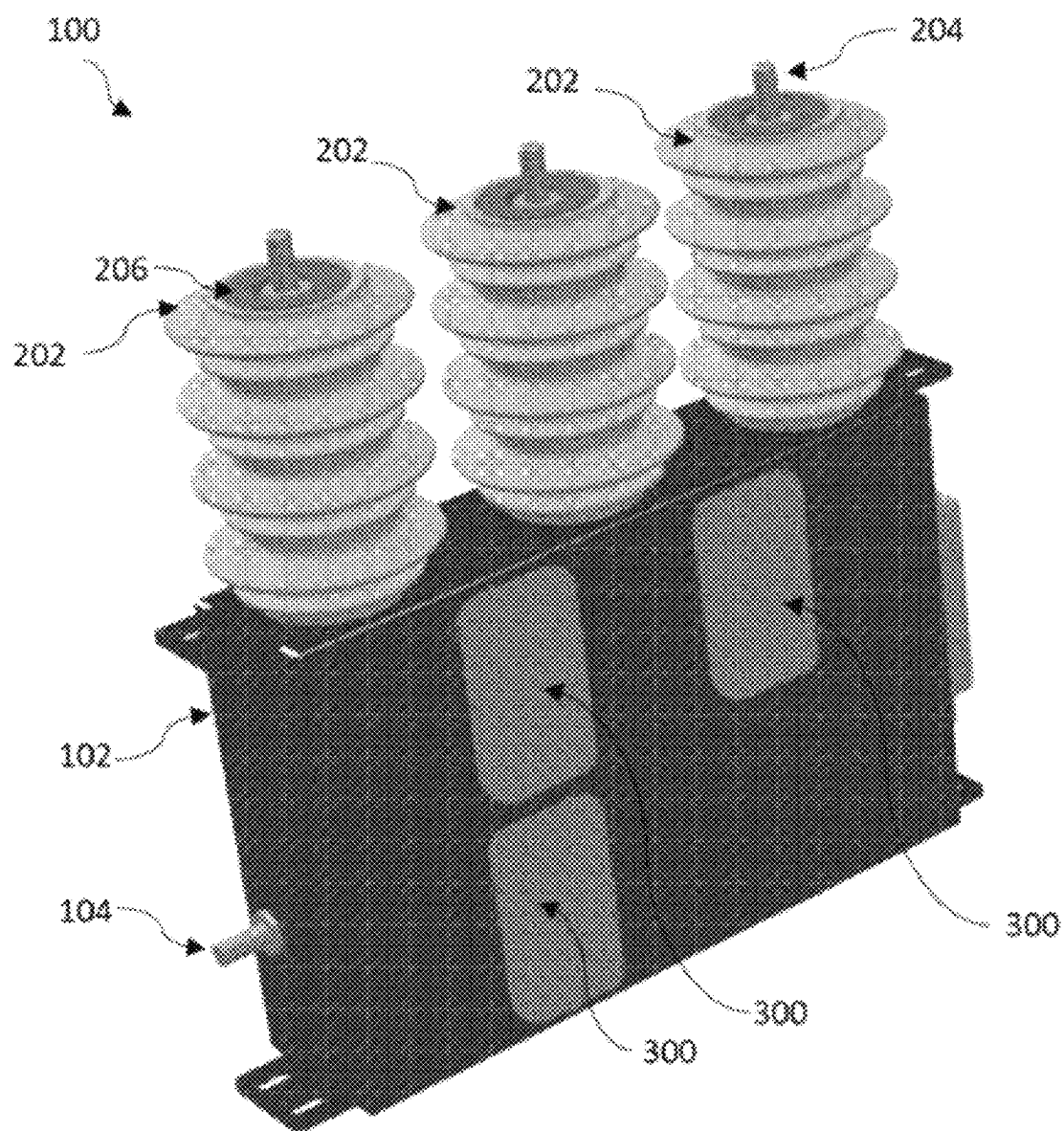


FIG. 1

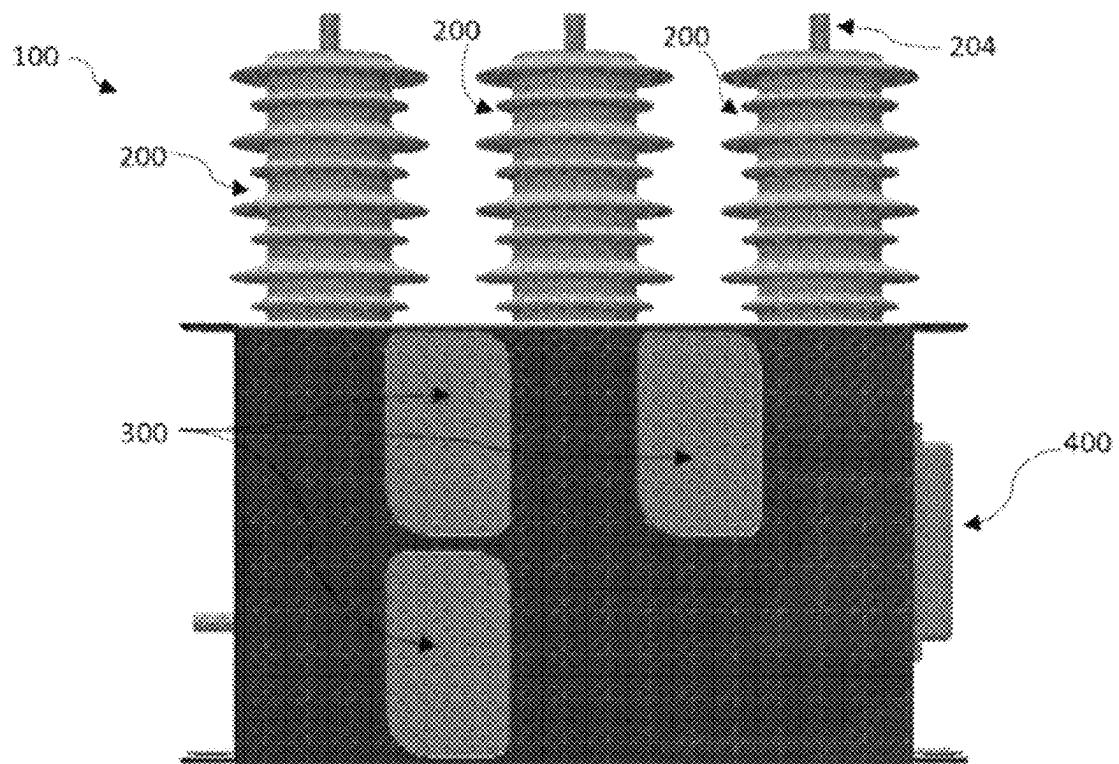


FIG. 2

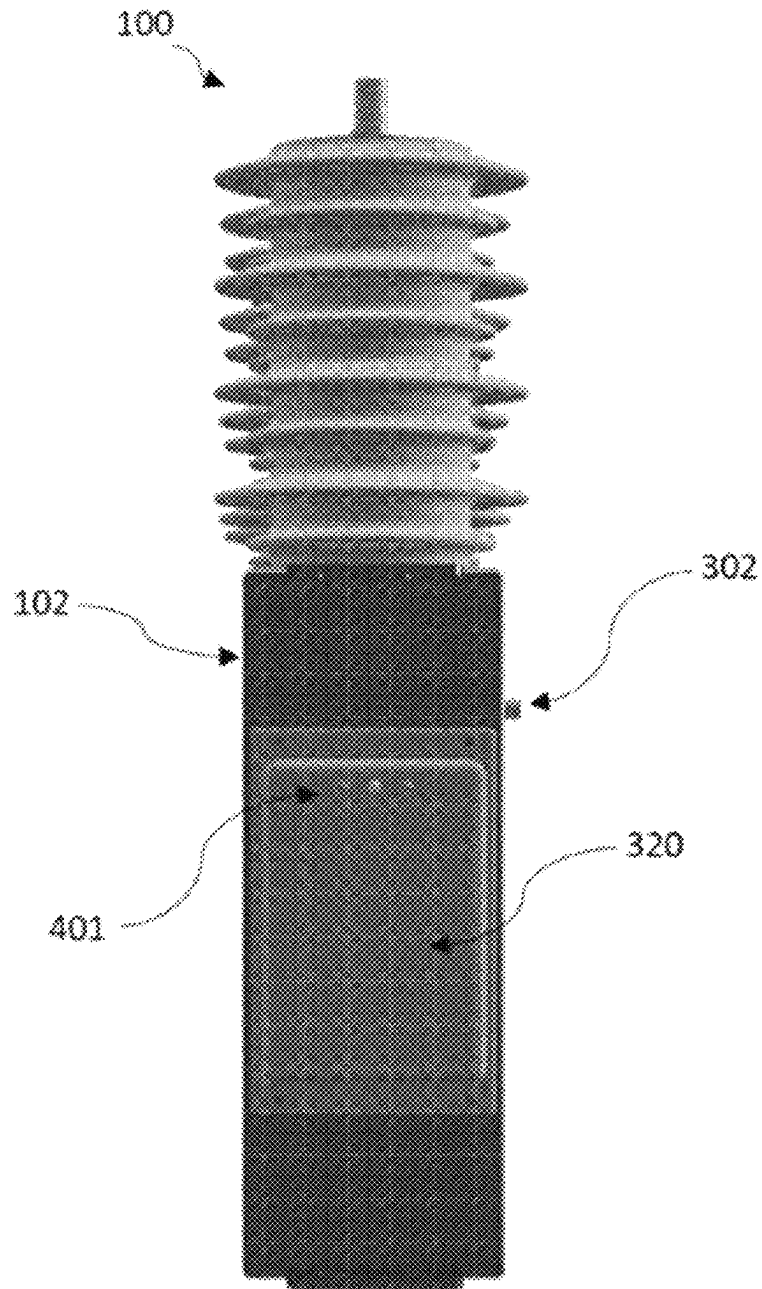


FIG. 3

200

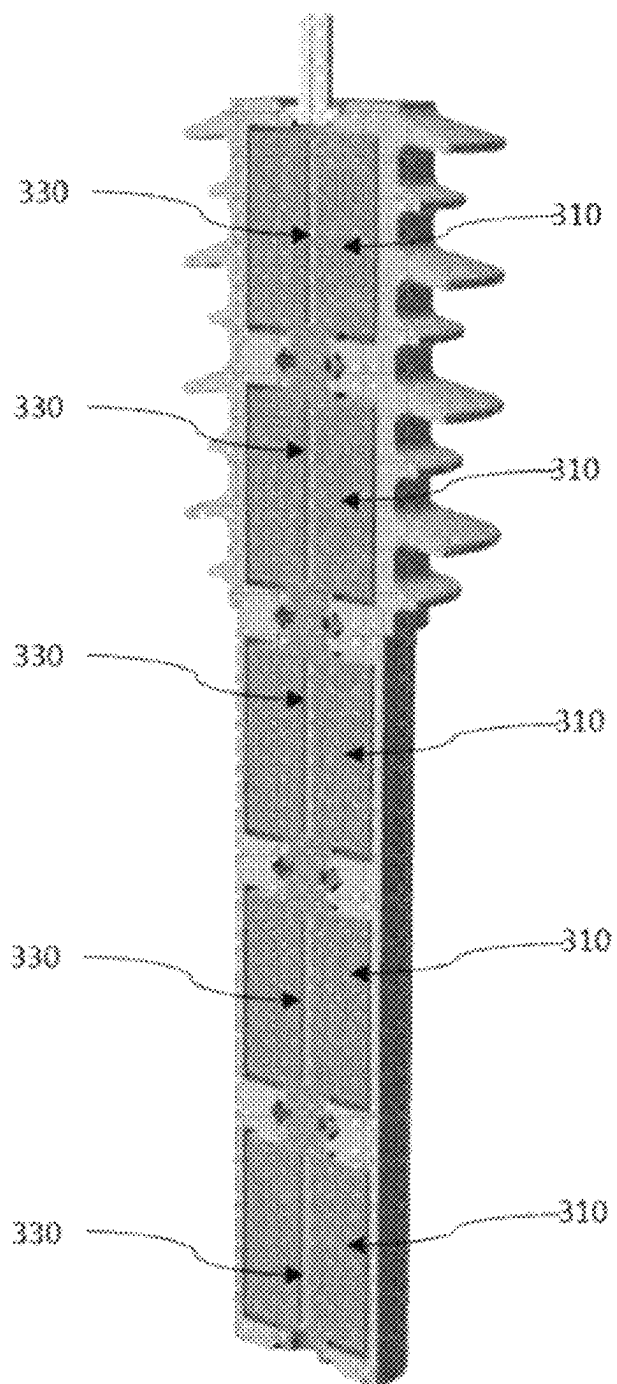


FIG. 4

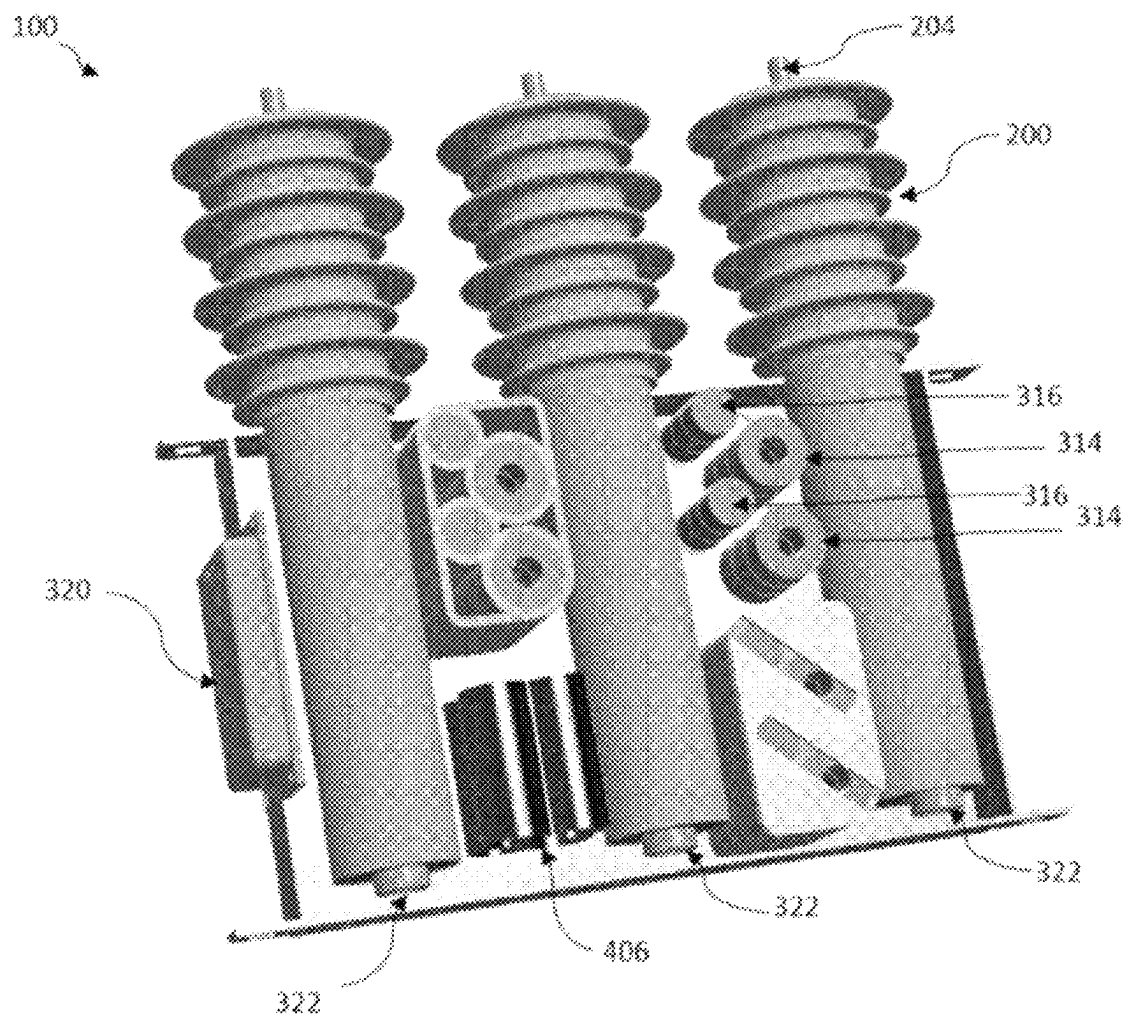
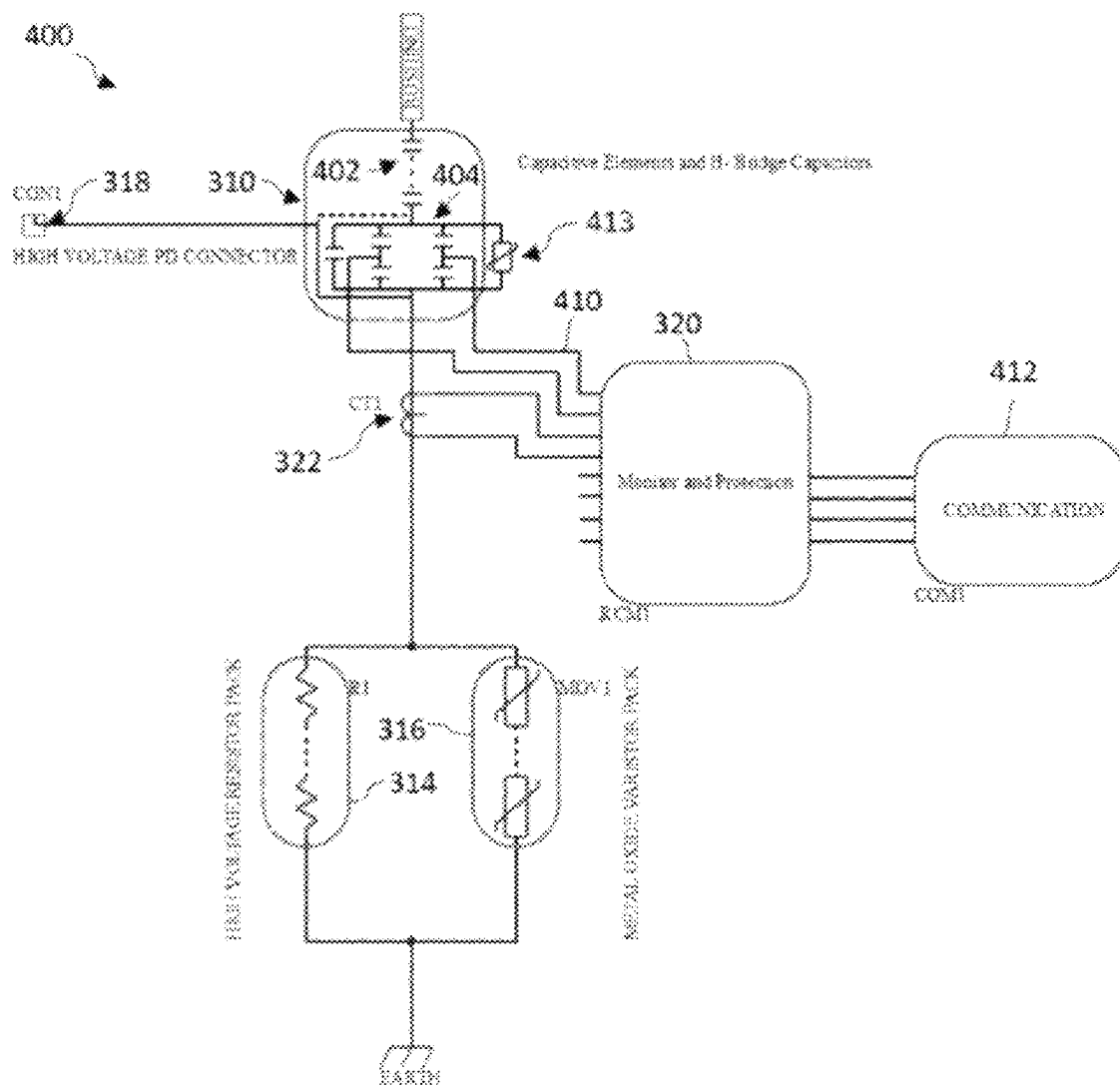


FIG. 5



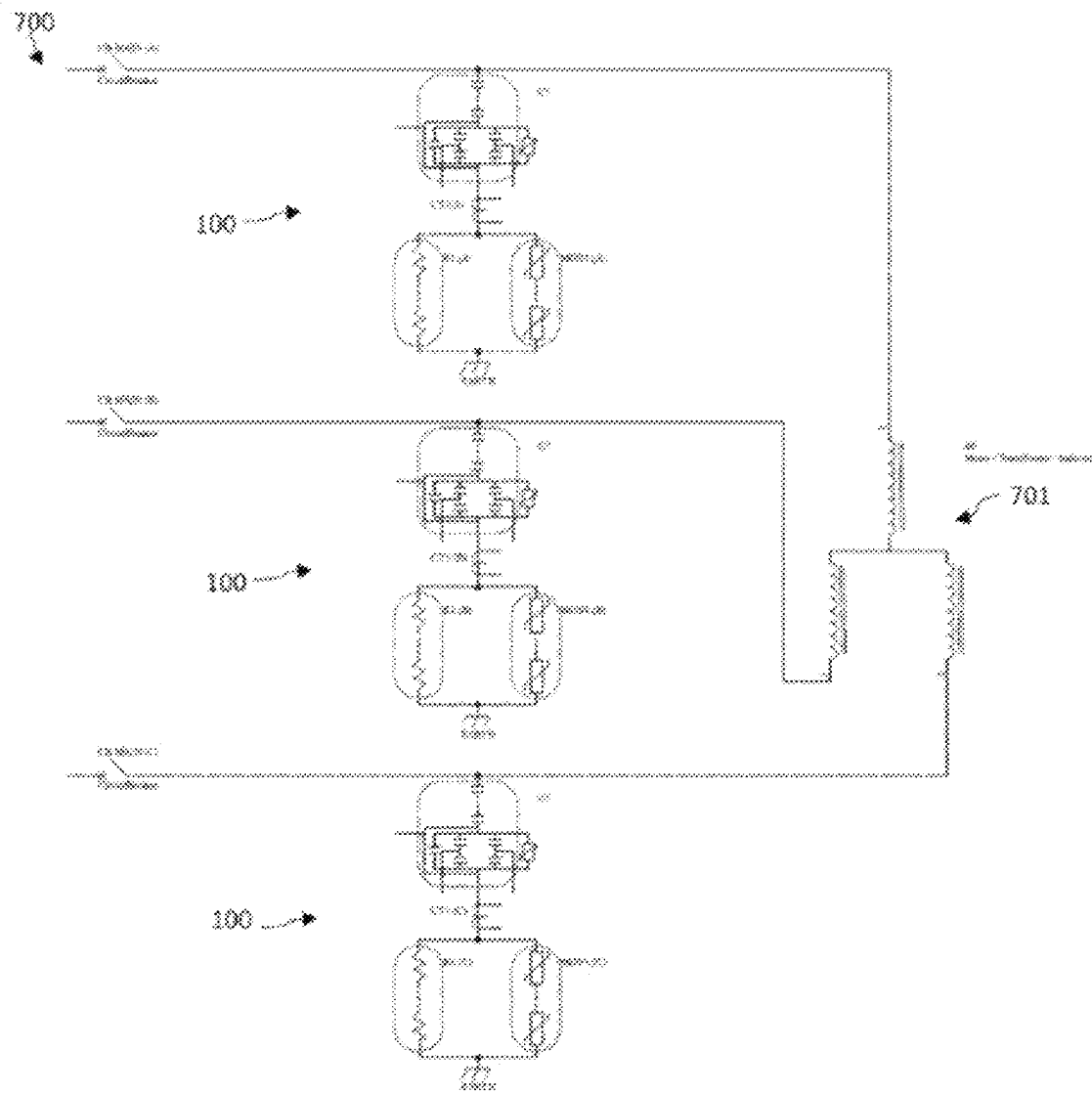


FIG. 7

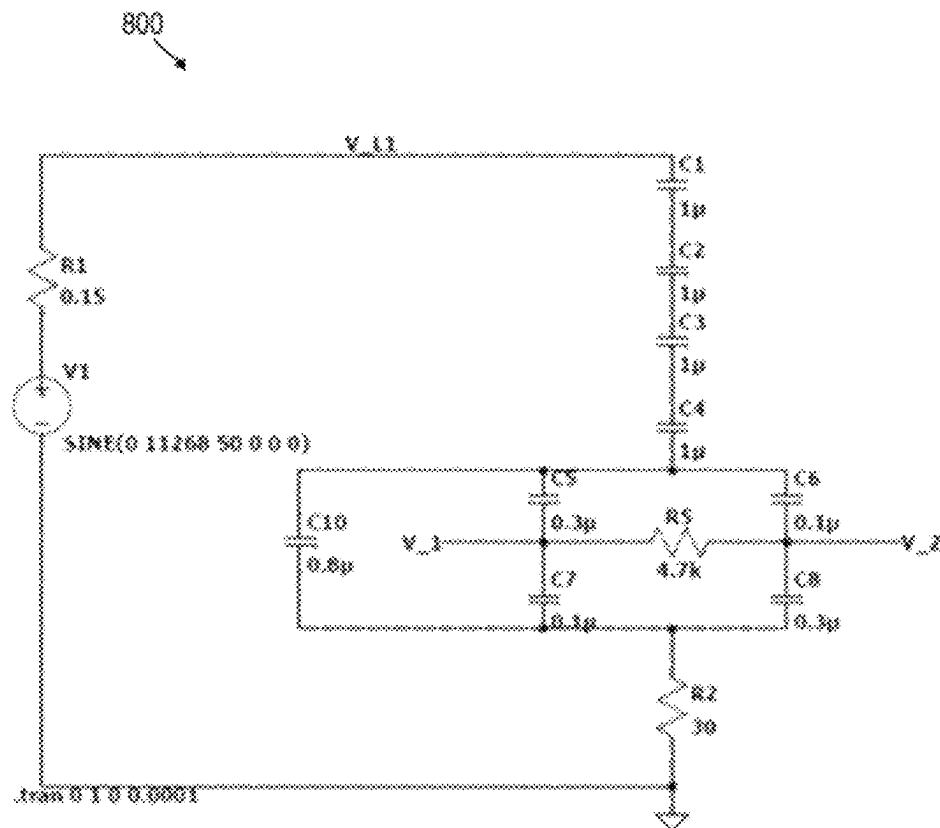


FIG. 8



FIG. 9

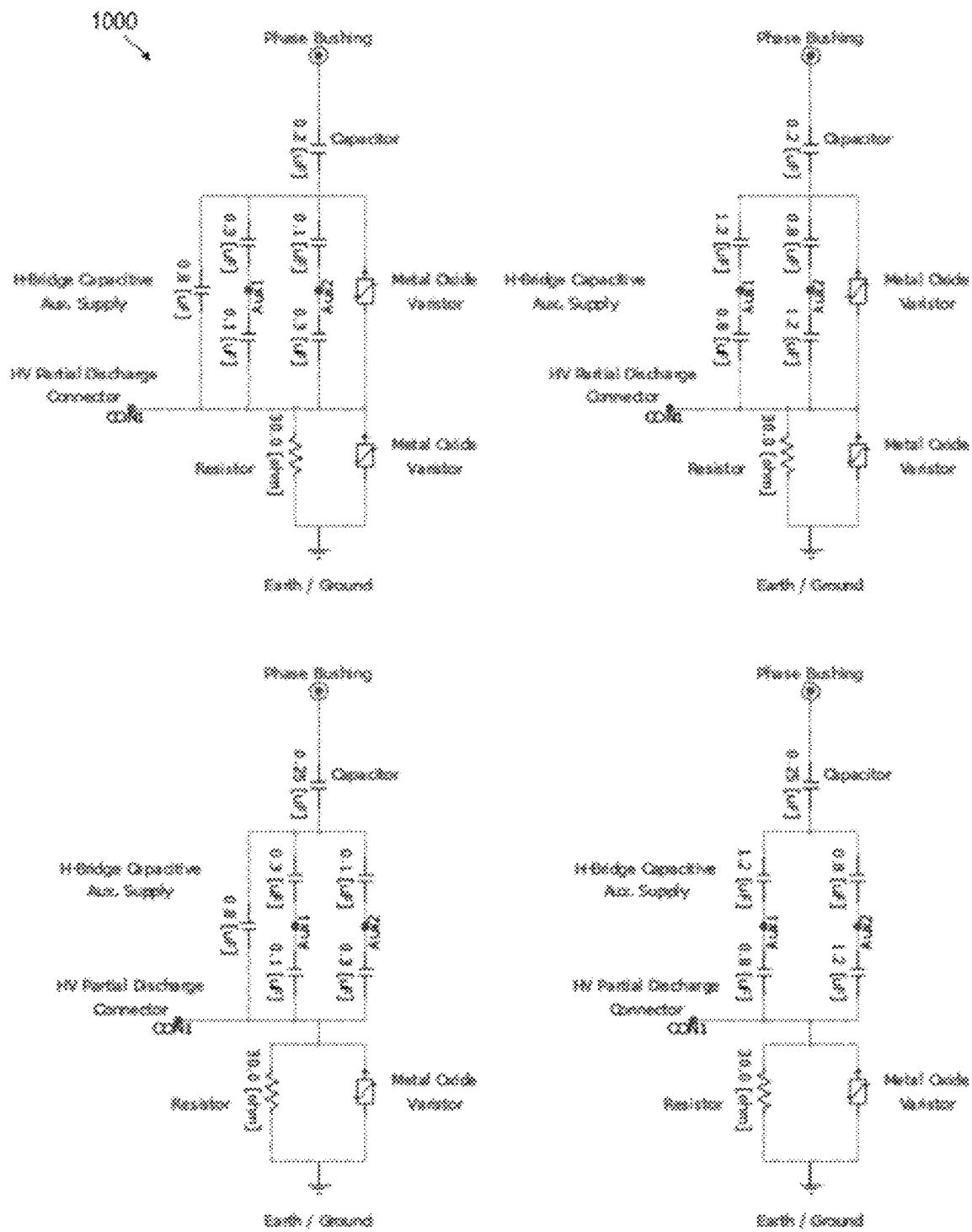


FIG. 10

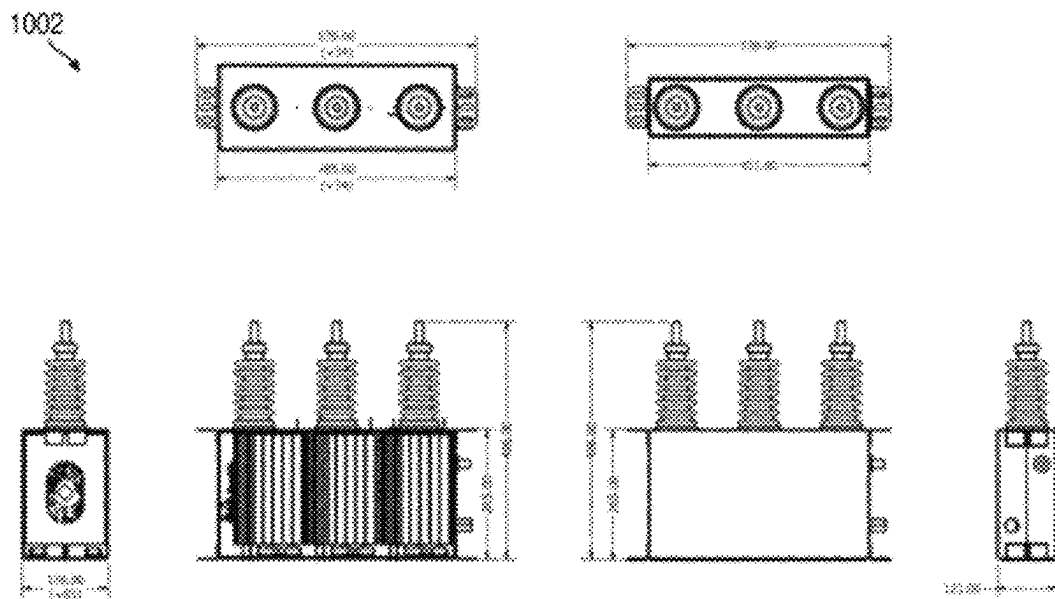


FIG. 11

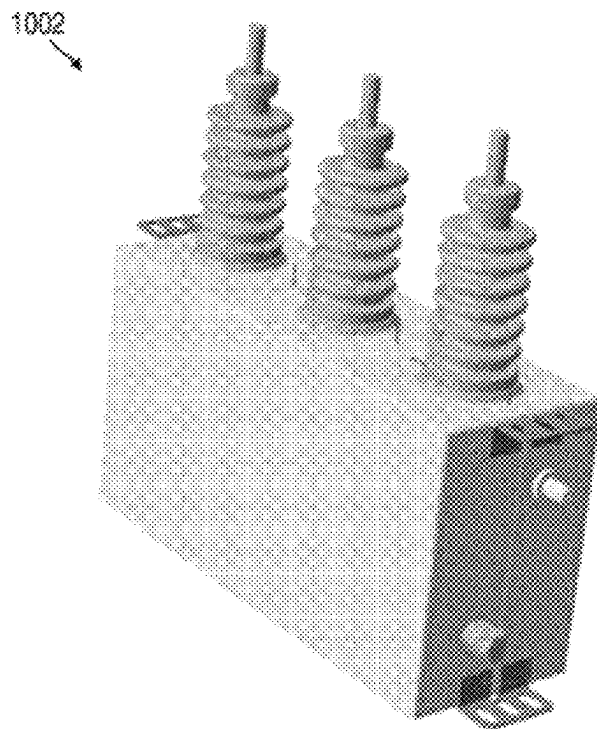


FIG. 12

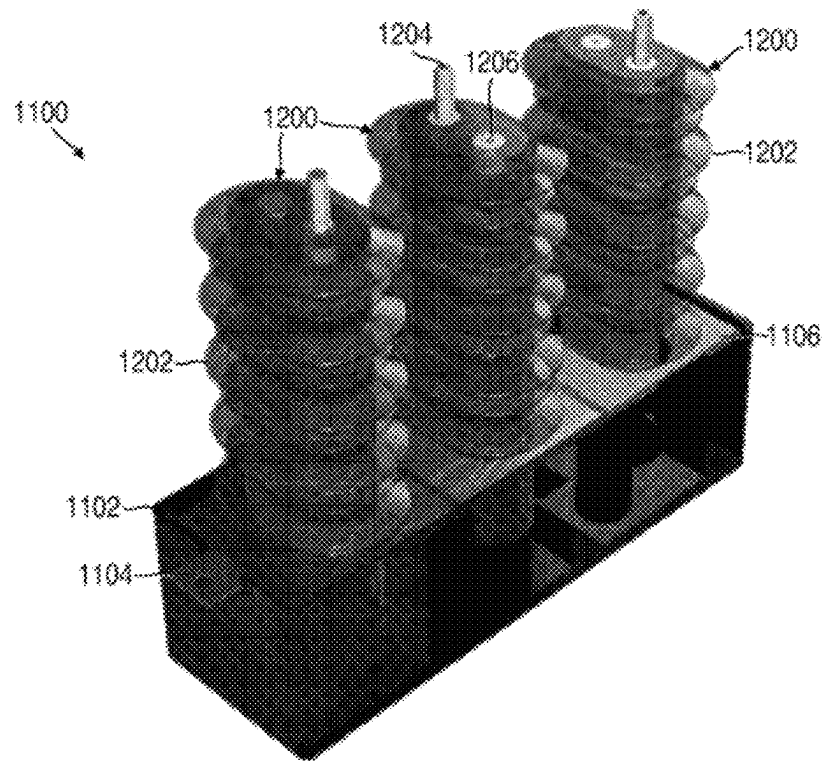


FIG. 13

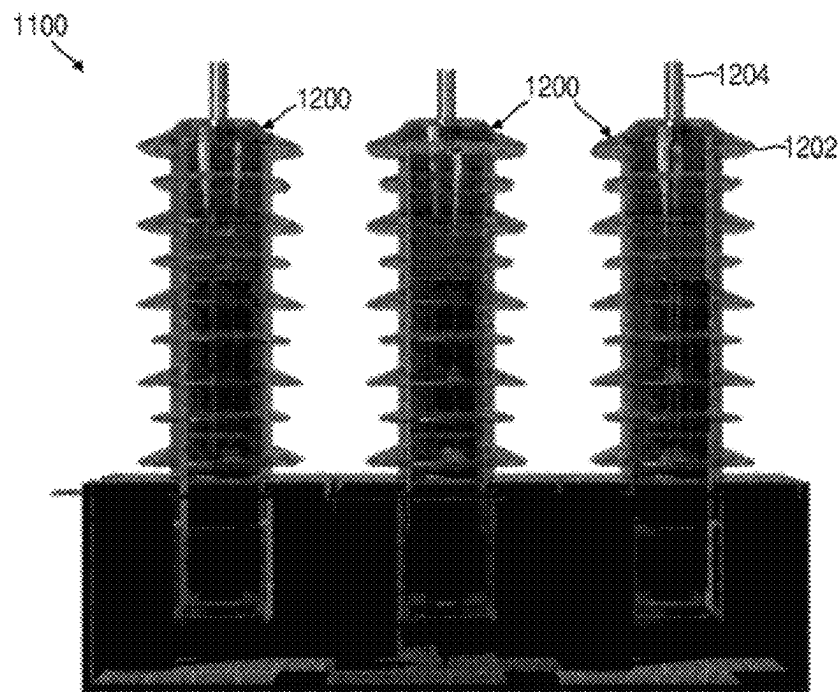


FIG. 14

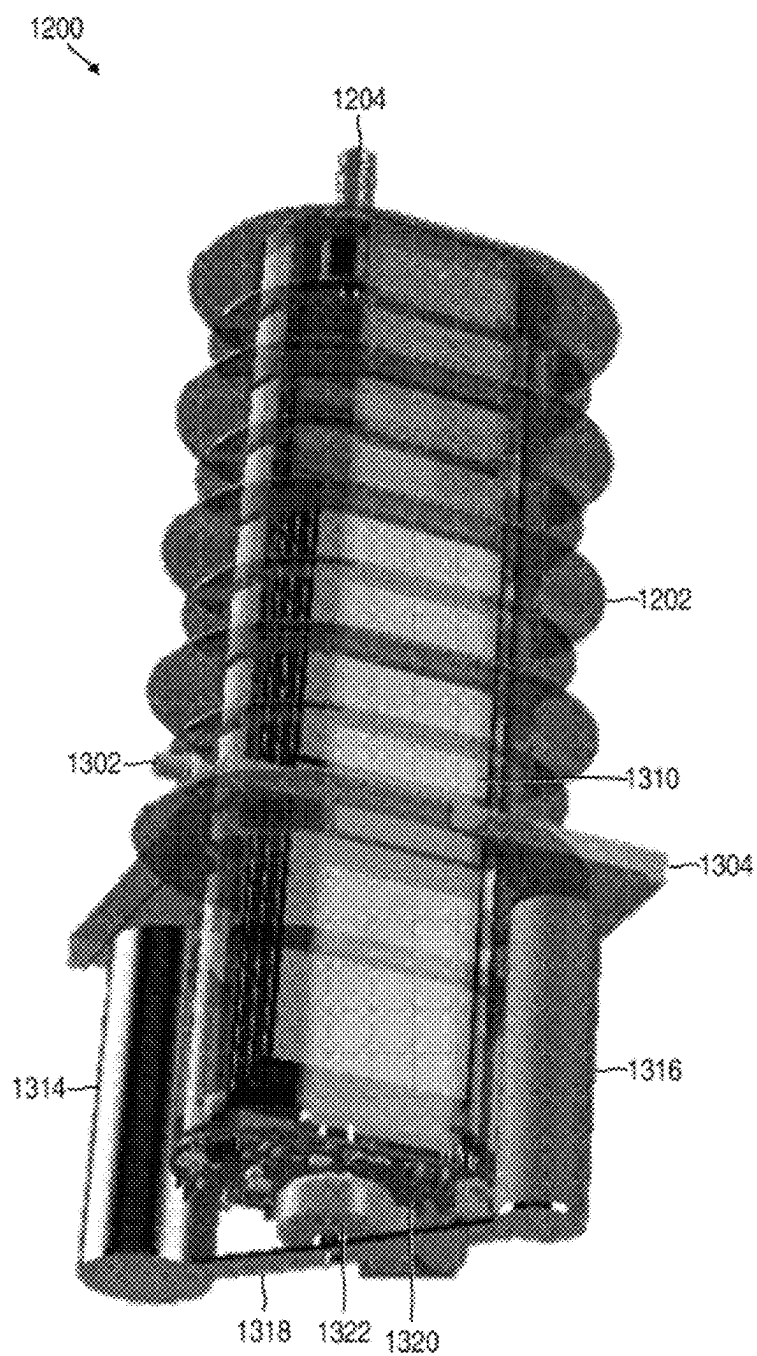


FIG. 15

1

ELECTRICAL SURGE SUPPRESSOR**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a National Stage entry of International Application No. PCT/IB2022/053097, filed Apr. 4, 2022, which claims priority to South African Patent No. 2021/02395, filed Apr. 13, 2021.

FIELD OF INVENTION

This invention relates to electrical power safety and equipment/asset protection and more specifically to a surge suppressor for low- to medium-voltage electrical equipment.

BACKGROUND OF INVENTION

The Inventor invented an electrical surge suppressor as disclosed in South African Patent No. 82/05995. This surge suppressor is effective at suppressing surges during switching in medium or high voltage switch controlled, cable connected, electric motor, transformer, or generator circuits.

A related, subsequent invention was disclosed in South African Patent No. 2011/04748, whereby the configuration of the capacitive and resistive (linear and non-linear) components was swapped, and modular components utilised. By providing a capacitive connection to the phase, this component serves as a coupling mechanism for partial discharge monitoring.

The Inventor now desires an improved surge suppressor providing even more modular components and updated monitoring capabilities.

SUMMARY OF INVENTION

The invention provides an electrical surge suppressor configured to be connected between earth and a phase of a switch controlled, AC electric load circuit, the surge suppressor including:

- a capacitive component incorporated inside a housing and immersed in, or filled with, oil;
- a linear resistive component having a linear V-I characteristic and being connected to the earth;
- a non-linear resistive component connected or connectable in parallel with the linear resistive component and having a pre-determined knee-point voltage value;
- a monitoring circuit configured to measure at least one operating characteristic of the linear and/or non-linear resistive and capacitive components; and
- an auxiliary power supply integrated within the capacitive component to provide an auxiliary supply voltage to the monitoring circuit,

wherein:

- the capacitive component is connected in series with the parallel connected linear and non-linear resistive components and connected to a phase contact;
- the phase contact is connected or connectable to the load circuit;
- the capacitive component is configured to provide a high impedance connection to the linear and non-linear resistive components, at nominal power supply frequencies and, during switching transients, to provide a low impedance to a combined resistive impedance of the linear and non-linear resistive components effectively into the load circuit at frequencies associated with wavefront rise times of up to 5 μ s (microseconds);

2

the capacitive component is configured to increase the rise time of a portion of a steep-fronted surge which exceeds a knee-point voltage value of the non-linear resistive component; and

- an external contact is arranged between the linear resistive component and the capacitive component.

The capacitive component may be a main capacitive component and auxiliary power supply may be provided by a secondary capacitive component connected to the main capacitive component. The auxiliary supply voltage may eliminate the need for an external high voltage transformer.

The auxiliary power supply may be configured such that under steady state conditions, a capacitive reactance of the secondary capacitive component is increased thereby reducing power dissipation and providing the capability to handle higher total harmonic voltage distortion levels. The auxiliary power supply may be configured such that under transient conditions, an impedance of the secondary capacitive component is effectively limited when the non-linear resistive component reaches or exceeds the knee-point voltage, reducing the total capacitive reactance and thereby increasing the rise time of the portion of the steep-fronted surge. Thus, the RC time constant ($\tau=RC$) may be increased due to the increase in capacitance and increasing the rise time.

The external contact may be used for testing components of the surge suppressor. In the context of this specification, testing may include type testing, routine testing and/or maintenance testing. By applying a testing apparatus across the external contact and earth, the linear resistive component may be tested. In this fashion, the influence of the capacitive component may be excluded when testing the linear resistive component. The testing apparatus may be operable to apply a DC test voltage across the linear resistive component, thereby to load the resistive component to test its rated power and then optionally to assess thermal impact on the capacitive component.

Similarly, by applying the testing apparatus across the phase contact and the external contact, the capacitive component may be tested. In this fashion, the influence of the linear resistive component may be excluded when testing the capacitive component. The testing apparatus may be operable to apply and AC or DC test voltage across the capacitive component.

In addition, an external monitoring device such as a Partial Discharge (PD) monitor, may be connected to the surge suppressor, thereby assessing, and predicting imminent failures of motors, transformer, and/or generators.

The monitoring circuit may be configured to measure at least one operating characteristic in the form of current, voltage, and/or temperature. The monitoring circuit may include one or more sensors (e.g., temperature sensor, voltage sensor, current sensor, etc.). The monitoring circuit may be configured to provide a protection function in that if the at least one operating parameter violates an operating threshold, the monitoring circuit implements a protective action. The protective action may include disconnecting one or more components, providing a short or open circuit, triggering an alarm, etc.

The monitoring circuit may include a communications device. The communications device may be configured to communicate measured operating characteristics and status to a recipient. The recipient may be one or more of smart-phone/tablet device, a human machine interface (HMI), and/or a cloud/web-based interface.

The monitoring circuit may include a display component, e.g., an LED or other light emitter or LCD/OLED. The display component may be configured to provide an indi-

cation (e.g., a visual indication) of one of the operating characteristics and/or of a violated operating threshold (e.g., a warning, alarm, or error/trip condition).

The auxiliary power supply may be a low voltage power supply below 1000 Vac. The monitoring circuit may be powered by the low voltage power supply.

The capacitive component may be provided by a capacitor pack comprising a plurality of capacitors. The capacitors may be in series, in parallel, or a combination of series and parallel. The capacitor pack may comprise an H-bridge configuration. The H-bridge configuration may provide points from which to provide the auxiliary power supply.

The bushing may be transparent/translucent or have a transparent/translucent window, to enable viewing an interior, e.g., to view a level or state of the oil.

The electrical surge suppressor may be modular, allowing for modular exchange of its components. An advantage of this modular configuration is that a standard or uniform base module may be manufactured. The electrical surge suppressor may include a variety of interchangeable modules, each with its non-linear resistive component being rated at different voltage levels, for example between 3.3 kV and 15 kV, thereby to adjust or modify the rating of the surge suppressor as a whole. Also, the non-linear resistive component may be tested independently of the other components by applying a testing apparatus across its contacts when the interchangeable module has been removed from the base module. This also allows for a specific non-linear resistive component to be selected so that its knee-point voltage characteristics and energy rating are suitable for a specific application.

The surge suppressor may be modular, and the housing may be embodied by a bushing, wherein the capacitive component, the linear resistive component, and the non-linear resistive component are mounted in or to the bushing. The capacitive component may be provided inside the bushing and the bushing may be filled with the oil, thereby to immerse the capacitor in the oil. The linear resistive component, the non-linear resistive, and the monitoring circuit may be mounted to an outside of the bushing, thus being separated from the oil inside the bushing.

The linear and/or non-linear resistive components may be modular and replaceable. For example, if the linear resistive component fails, it may be removed and replaced with a new one, without needing to discard the whole surge suppressor.

The invention extends to a surge suppressor assembly including at least one electrical surge suppressor as defined above accommodated in a base unit. The surge suppressor assembly may include three electrical surge suppressors accommodated in the base unit, for a three-phase power system.

The base unit may include a common earth connection. The surge suppressor(s) may be electrically connected to the base unit, thereby to be electrically connected to the common earth connection.

The invention extends to a surge suppressor kit for assembling into a surge suppressor assembly as defined above, in which a plurality of surge suppressors are provided with different ratings or characteristics, such that the surge suppressor assembly can be constituted using the base and one or more surge suppressors selected from plurality of surge suppressors provided with the different ratings or characteristics.

BRIEF DESCRIPTION OF DRAWINGS

The invention will now be further described, by way of example, with reference to the accompanying diagrammatic drawings.

In the drawings:

FIG. 1 shows a three-dimensional view of a surge suppressor assembly, in accordance with the invention;

FIG. 2 shows a side view of the surge suppressor assembly of FIG. 1;

FIG. 3 shows a side view of the surge suppressor assembly of FIG. 1;

FIG. 4 shows modular surge capacitors of the assembly of the surge suppressor of FIG. 1;

FIG. 5 shows a three-dimensional view of an electrical surge suppressor assembly of FIG. 1;

FIG. 6 shows a circuit diagram of the surge suppressor of FIG. 1;

FIG. 7 shows a circuit diagram of the typical application of the surge suppressor assembly of FIG. 1;

FIG. 8 shows a circuit diagram of an H-bridge configuration of the surge suppressor assembly of FIG. 1;

FIG. 9 shows a graph of a simulated unbalanced power supply provided by the H-bridge of FIG. 8;

FIG. 10 shows circuit diagrams of alternative H-bridge configurations of the surge suppressor assembly of FIG. 1;

FIG. 11 shows various schematic views of an alternative surge suppressor of FIG. 1 in a metal enclosure and ceramic bushings;

FIG. 12 shows a three-dimensional view of the surge suppressor of FIG. 11;

FIG. 13 shows a three-dimensional drawing of a surge suppressor assembly, in accordance with the invention;

FIG. 14 shows a side view of the surge suppressor assembly of FIG. 13; and

FIG. 15 shows a three-dimensional view of an electrical surge suppressor forming part of the surge suppressor assembly of FIG. 13.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENT

The following description of an example embodiment of the invention is provided as an enabling teaching of the invention. Those skilled in the relevant art will recognise that changes can be made to the example embodiment described, while still attaining the beneficial results of the present invention. It will also be apparent that some of the desired benefits of the present invention can be attained by selecting some of the features of the example embodiment without utilising other features. Accordingly, those skilled in the art will recognise that modifications and adaptations to the example embodiment are possible and can even be desirable in certain circumstances and are a part of the present invention. Thus, the following description of the example embodiment is provided as illustrative of the principles of the present invention and not a limitation thereof.

FIGS. 1-2 illustrate a surge suppressor **100**, in accordance with the invention. The surge suppressor **100** is intended for three-phase operation (although other variants may be practicable) and accordingly it comprises three high-voltage bushing units **200** and component packs **300** mounted to a base unit **102**.

The outward appearance of the assembly of the surge suppressor **100**, that is, of the base unit **102**, the bushing units **200**, and component packs **300** may vary based on operational conditions or design criteria. The base unit **102** is rectangular and defines sockets to accommodate the bushing units **200** and component packs **300**. The base unit **102** has a common earth terminal **104**. Earth terminals (not illustrated) of each of the component packs **300** are also electrically connected to the common earth terminal **104** by

means of reliable conductive mechanical fasteners, e.g., cables, lugs, bolts, terminals, spring force, etc.

Each bushings unit **200** comprises a high voltage bushing **202** having a shed-profile for required clearance and creepage distances. The bushing **202** is hollow and optionally translucent, providing at least limited visibility into its interior. A phase (or live) terminal **204**, in the form of an upwardly projecting threaded stud **204**, is provided at a top of the bushing **202** for connection to a phase line. An oil filler plug **206** is provided in an aperture at the top of the bushing **202** or individual capacitor element (not illustrated), to access an interior, e.g., to fill or access the oil therein. The oil filler plug **206** may be used to service, drain, or refill oil within the bushing **202**. However, oil-filled bushings per se are known, so these details are not germane to the invention. The oil may be filled, used, configured, etc. using conventional techniques. The bushing **202** may be of a non-ceramic material, e.g., a polymer, fibre glass reinforced material with polymer shed profiles.

FIG. **3** illustrates the surge suppressor, **100** with a connection point **302** for an external monitoring device in the form of a partial discharge monitoring device provided at a side of the base unit **102**. A monitoring circuit **320** may be located at a side of the base unit **102** with indication of the status of the surge suppressor **100** (including the communications interface).

FIG. **4** illustrates a cross-section of the bushing unit **200** with a capacitive component **310** provided inside the bushing **202** and is therefore immersed in, or filled with, oil. The capacitive component **310** could be one very large capacitor but, in this example, it is in the form of a capacitor pack **310** comprising a plurality of capacitors (see FIGS. **4-6** for further details). The capacitive components **310** may contain discharge resistors **330** in order to safely discharge the voltage stored in the capacitive element when ac voltage supply is switched off.

FIG. **5** illustrates the surge suppressor **100** with a linear resistive component, in the form of a high voltage resistor pack **314**, having a linear V-I characteristic and being connected to the earth, separated from the oil in the bushing **200**. Similarly, the surge suppressor **100** has a non-linear resistive component, in the form of a metal oxide varistor pack (or MOV pack for short) **316**, connected or connectable in parallel with the resistor pack **314** and having a pre-determined knee-point voltage value.

The resistor pack **314** and the MOV pack **316** are connected in parallel, while the capacitive component **310** is connected in series with the parallel connected resistor and MOV packs **314**, **316**. The bushing unit **200** containing the capacitive component **310** (FIG. **4**) is connected to the phase contact **204** and is configured to provide a high impedance connection to the resistive and MOV packs **314**, **316** at nominal power supply frequencies (50-60 Hz), and to couple a combined resistive impedance of the resistive and MOV packs **314**, **316** effectively into the load circuit at frequencies associated with wavefront rise times of up to five microseconds. The capacitive component **310** (FIG. **4**) in the bushing **200** is configured to increase the rise time of a portion of a steep-fronted surge which exceeds a knee-point voltage value of the MOV pack **316**.

The monitoring circuit **320** is provided and is configured to measure at least one operating characteristic of the linear and/or non-linear resistive components **314**, **316** or other components. The monitoring circuit **320** is configured to provide monitoring and/or protection of the individual components including system parameters (e.g., the resistive and MOV packs **314**, **316**, system voltage etc.) of the surge

suppressor **100**. The monitoring circuit **320** interfaces with a current transformer (CT) **322** and includes various other sensors (temperature, pressure, humidity etc.) configured to measure parameters providing optimum protection for the surge suppressor **100**. A section of the H-bridge **406** is configured to provide auxiliary power (refer to FIG. **6**).

A communication device is connected to, or integrated with, the monitoring circuit **320** and is configured to transmit the measured parameters to external recipients (e.g., cloud-based monitoring, predictive and maintenance algorithms).

Referring now to FIG. **6**, a circuit diagram **400** of the surge suppressor **100** with a single bushing unit **202** is illustrated. The same reference numerals in the FIGS refer to the same or similar parts.

FIG. **6** illustrates a composition of the capacitive component **310** which comprises several capacitors. More specifically, the capacitive component **310** comprises multiple high-voltage capacitors **402** connected in series in order to be able to withstand high system voltages that the capacitive component **310** will be exposed to during normal operation. Variations in the number of capacitors **402** connected in series and the layers and thickness of film and foil used is dependent on a particular design based on the required safety factors and margins. As mentioned previously, the capacitive component **310** is submerged in oil in order to increase the capacitance (the oil serves as a dielectric) and provide additional insulation resistance, as well as assisting with heat transfer/dissipation.

Further, the capacitive component **310** comprises a capacitive H-bridge **404** which, in turn, comprises plural capacitors arranged in parallel. The H-bridge **404** is configured to an unbalanced voltage between centre points of the bridge to provide an auxiliary power point **410** to which auxiliary equipment (e.g., the monitoring circuit **320**) can be connected and powered. This configuration may eliminate a requirement for high voltage step down transformers or the need for wide range power supplies to power auxiliary equipment. The high-voltage capacitors **402** and/or the H-bridge **404** can be contained in separate units, or a single oil filled container.

There may be various capacitive component **310** design options:

The high-voltage capacitors **402** may all be in their own oil-filled container and then provided in series with the H-bridge **404** being in its own oil-filled container (for a total of two separate oil-filled containers);

Each high-voltage capacitor **402** may be in its own oil-filled container which are then connected in series and, in turn, provided in series with the H-bridge **404** being in its own oil-filled container (for a total of three or more separate oil-filled containers); or

All capacitors **402**, **404** may be in a single oil-filled container (for a total of one oil-filled container).

There may be secondary MOVs **413** connected in parallel with the H-Bridge **404** capacitive element in order to effectively remove the capacitor function of the H-Bridge **404** during fast rising transient conditions in order to increase the capacitance of the surge suppressor **100**.

The resistive pack **314** may comprise a plurality of individual resistors connected in series/parallel configurations depending on the power rating required for the suppressor design. The MOV pack **316** may comprise multiple metal oxide varistors placed in series and rated depending on the required clamping voltage and power rating of the suppressor design. The CT (current transformer) **322** pro-

vides a signal to the monitoring circuit **320** to derive the required parameters to protect the components of the surge suppressor **100**.

The external contact **318**, which may be considered a high voltage partial discharge connector, provides a connection point for external monitoring equipment to be connected (typically partial discharge monitoring) as well as access to the individual components (e.g., the resistive and MOV packs **314**, **316**) in the surge suppressor **100**. The external contact **318** provides direct access to the capacitors **402**, **404** coupled to the supply lines of the equipment to be monitored.

The auxiliary power supply connection **410** is provided by "tapping" off from the capacitors in the H-bridge **404**. The auxiliary connection **410** is used to power the monitoring circuit **320**.

FIG. **7** illustrates a circuit diagram **700** of the surge suppressor **100** with all three bushing units **200** for a three-phase application (e.g., a three-phase motor **701**).

FIGS. **8-9** illustrate an H-bridge configuration **800** of the surge suppressor **100** configured to provide a power supply between points **V_1** and **V_2** having a waveform **900**.

FIG. **10** illustrates an alternative H-bridge configuration **1000** with various configuration options.

FIGS. **11-12** illustrate an alternative configuration **1002** of the surge suppressor **100**, containing all the components in a current form factor utilising a metal enclosure and ceramic bushings, with communications capability provided via a fibre optic interface. The components may not be modular/replaceable.

FIGS. **13-15** illustrates a different embodiment in the form of a surge suppressor assembly **1100**, in accordance with the invention. The assembly **1100** is intended for three-phase operation (although other variants may be practicable) and accordingly it comprises three electrical surge suppressors **1200** mounted in a base unit **1102**. Several features of the assembly **1100** may be similar or identical to those of the surge suppressor **100**.

The outward appearance of the assembly **1100**, that is, of the base unit **1102** and the surge suppressors **1200** may vary based on operational conditions or design criteria. The base unit **1102** is rectangular and defines three sockets to accommodate the three surge suppressors **1200**. The base unit **1102** has a common earth terminal **1104** connected to a common earth plate **1106**. Earth terminals (not illustrated) of each of the surge suppressors **1200** are also electrically connected to the common earth plate **1106** by means of reliable conductive mechanical fasteners, e.g., cables, lugs, bolts, terminals, spring force, etc.

Each surge suppressor **1200** comprises a high voltage bushing **1202** having a shed-profile for required clearance and creepage distances. The bushing **1202** is hollow and potentially translucent, providing at least limited visibility into its interior. A phase (or live) terminal **1204**, in the form of an upwardly projecting threaded lug **1204**, is provided at a top of the bushing **1202** for connection to a phase line. An oil filler plug **1206** is provided in an aperture at the top of the bushing **1202**, to access an interior, e.g., to fill or access the oil therein. The oil filler plug **1206** may be used to service, drain, or refill oil within the bushing **1202**. However, oil-filled bushings per se are known, so these details are not germane to the invention. The oil may be filled, used, configured, etc. using conventional techniques. The bushing **1202** may be of a non-ceramic material, e.g., a polymer, fibre glass reinforced material with polymer shed profiles.

FIG. **15** illustrates the surge suppressor **1200** in more detail. A connection point for a partial discharge monitoring

device **1302** is provided at a side of the bushing **1202**. The surge suppressor **1200** has a square collar **1304** towards a mid-bottom region of the bushing **1202** to locate the bushing **1202** snugly in the base unit **1102**.

A capacitive component **1310** is provided inside the bushings **1202** and is therefore immersed in, or filled with, oil. The capacitive component **1310** could be one large capacitor but, in this example, it is in the form of a capacitor pack **1310** comprising a plurality of capacitors.

The surge suppressor **1200** has a linear resistive component, in the form of a high voltage resistor pack **1314**, having a linear V-I characteristic and being connected to the earth, separated from the oil in the bushing **1202**. Similarly, the surge suppressor **1200** has a non-linear resistive component, in the form of a metal oxide varistor pack (or MOV pack for short) **1316**, connected or connectable in parallel with the resistor pack **1314** and having a pre-determined knee-point voltage value. The MOV pack **1316** is also separated from the oil of the capacitive component **1310** inside the bushing **1202**.

The resistor pack **1314** and the MOV pack **1316** are connected in parallel, while the capacitive component **1310** is connected in series with the parallel connected resistor and MOV packs **1314**, **1316**. An external contact **1318** is arranged between the resistive and MOV packs **1314**, **1316** and the capacitive component **1310**. The capacitive component **1310** is connected to the phase contact **1204** and is configured to provide a high impedance connection to the resistive and MOV packs **1314** and **1316** at nominal power supply frequencies (50-60 Hz), and to couple a combined resistive impedance of the resistive and MOV packs **1314**, **1316** effectively into the load circuit at frequencies associated with wavefront rise times of up to five microseconds. The capacitive component **1310** is configured to increase the rise time of a portion of a steep-fronted surge which exceeds a knee-point voltage value of the MOV pack **1316**.

A monitoring circuit **1320** is provided and is configured to measure at least one operating characteristic of the linear and/or non-linear resistive components **1314**, **1316** or other components. The monitoring circuit **1320** is configured to provide monitoring and/or protection of the individual components including system parameters (e.g., the resistive and MOV packs **1314**, **1316**, system voltage etc.) of the surge suppressor **1200**. The monitoring circuit **1320** interfaces with a current transformer (CT) **322** and various other sensors (temperature, pressure, humidity etc.) deriving measured parameters providing optimum protection for the surge suppressor **1200**.

A communication device is connected to, or integrated with, the monitoring circuit **1320** and is configured to transmit the measured parameters to external recipients (e.g., cloud-based monitoring, predictive and maintenance algorithms).

The surge suppressor assembly **1100** may include some or all of the features described in FIGS. **6-10**.

The Applicant believes that the invention as exemplified provides several advantages or overcomes various drawbacks of prior art systems, as follows:

Provides an auxiliary power supply that can be used to power monitoring and protection equipment as part of the capacitor design in an RC snubber/surge suppressor/power factor correction capacitors circuit. Deliberately creating an unbalance and tapping the auxiliary supply voltage from the unbalance.

Unbalanced capacitive power supply design protects the equipment connected by virtue of the capacitor impedance reduced in the event of switching transient surges.

Provides easier access to test individual components and a modular replacement of the individual parts.
 Eliminates exposing the heat generating components (resistive components) from the oil, thus preventing catastrophic failure and rupturing of the tank.
 Monitors the state of health of the various components in the RC snubber/surge suppressor/power factor correction capacitors (continuous or interval monitoring).
 Provides access to external partial discharge monitoring equipment utilising the internal capacitor as the PD (Partial Discharge) coupling device.
 Visual indication of the state of health of the RC snubber/surge suppressor/power factor correction capacitors (local or via remote communication and indication).
 RC Surge suppressor that is able to operate with continuous high Voltage harmonic distortion levels, if fitted with MOV **413**
 Modular components (Capacitors, Resistive Elements, Monitoring and Protection Device, Communications Modules).
 Provides and auxiliary power supply eliminating the need for high a voltage transformer by virtue of the capacitor pack configuration.
 Monitor state of health of the individual components and the surge suppressor as a system component.
 Components can be tested separately and replaced as and when required.
 Capacitor assembly utilising the internal space of the bushing reducing overall size and weight.

The invention claimed is:

1. An electrical surge suppressor configured to be connected between earth and a phase of a switch controlled, AC electric load circuit, the surge suppressor including:

a capacitive component incorporated inside a housing and immersed in, or filled with, oil;
 a linear resistive component having a linear V-I characteristic and being connected to the earth;
 a non-linear resistive component connected or connectable in parallel with the linear resistive component and having a pre-determined knee-point voltage value;
 a monitoring circuit configured to measure at least one operating characteristic of the linear and/or non-linear resistive and capacitive components; and
 an auxiliary power supply integrated within the capacitive component to provide an auxiliary supply voltage to the monitoring circuit,

wherein:

the capacitive component is connected in series with the parallel connected linear and non-linear resistive components and connected to a phase contact;

the phase contact is connected or connectable to the load circuit;

the capacitive component is configured to provide a high impedance connection to the linear and non-linear resistive components, at nominal power supply frequencies and, during switching transients, to provide a low impedance to a combined resistive impedance of the linear and non-linear resistive components effectively into the load circuit at frequencies associated with wavefront rise times of up to 5 μ s;

the capacitive component is configured to increase the rise time of a portion of a steep-fronted surge which exceeds a knee-point voltage value of the non-linear resistive component; and

an external contact is arranged between the linear resistive component and the capacitive component.

2. The surge suppressor as claimed in claim 1, in which the capacitive component is a main capacitive component and auxiliary power supply is provided by a secondary capacitive component connected to the main capacitive component.

3. The surge suppressor as claimed in claim 2, whereby the auxiliary power supply is configured such that (1) under steady state conditions, a capacitive reactance of the secondary capacitive component is increased thereby reducing power dissipation and providing the capability to handle higher total harmonic voltage distortion levels, but (2) under transient conditions, an impedance of the secondary capacitive component is effectively limited when the non-linear resistive component reaches or exceeds the knee-point voltage, reducing the total capacitive reactance and thereby increasing the rise time of the portion of the steep-fronted surge.

4. The surge suppressor as claimed in claim 1, in which the external contact is configured to be used for testing components of the surge suppressor, wherein:

by applying a testing apparatus across the external contact and earth, the linear resistive component can be tested; or

by applying a testing apparatus across the phase contact and the external contact, the capacitive component can be tested.

5. The surge suppressor as claimed in claim 1, which includes an external monitoring device configured to predict and assess imminent failures of motors, transformers, and/or generators.

6. The surge suppressor as claimed in claim 1, in which the monitoring circuit includes a sensor configured to measure at least one operating characteristic in the form of current, voltage, and/or temperature.

7. The surge suppressor as claimed in claim 6, in which the monitoring circuit is configured to provide a protection function in that if the at least one operating characteristic violates an operating threshold, the monitoring circuit implements a protective action, wherein the protective action includes one or more of disconnecting one or more components, providing a short or open circuit, or triggering an alert.

8. The surge suppressor as claimed in claim 6, in which the monitoring circuit includes a communications device which is configured to communicate at least the measured operating characteristic to a recipient.

9. The surge suppressor as claimed in claim 6, in which the monitoring circuit includes a display component which is configured to provide an indication of one of the operating characteristics and/or of a violated operating threshold.

10. The surge suppressor as claimed in claim 1, in which the auxiliary power supply is a low voltage power supply below 1000 Vac and in which the monitoring circuit is powered by the low voltage power supply.

11. The surge suppressor as claimed in claim 1, in which the capacitive component is provided by a capacitor pack comprising a plurality of capacitors.

12. The surge suppressor as claimed in claim 11, in which the plurality of capacitors are in an H-bridge configuration.

13. The surge suppressor as claimed in claim 12, in which the H-bridge configuration provides points from which to provide the auxiliary power supply.

14. The surge suppressor as claimed in claim 1, which is modular and in which the housing is embodied by a bushing, wherein the capacitive component, the linear resistive component, and the non-linear resistive component are mounted in or to the bushing.

11

15. The surge suppressor as claimed in claim **14**, in which: the capacitive component is provided inside the bushing and in which the bushing is filled with the oil, thereby to immerse the capacitor in the oil; and the linear resistive component, the non-linear resistive, and the monitoring circuit are mounted to an outside of the bushing, thus being separated from the oil inside the bushing.

16. The surge suppressor as claimed in claim **15**, in which the bushing is transparent or translucent or has a transparent or translucent window, to enable viewing an interior to view a level or state of the oil.

17. The surge suppressor as claimed in claim **14**, in which the linear and/or non-linear resistive components are modular, being replaceable or interchangeable without needing to replace a remainder of the surge suppressor.

18. A surge suppressor assembly including at least one electrical surge suppressor as claimed in claim **14** accommodated in a base unit.

12

19. The surge suppressor assembly as claimed in claim **18**, in which a plurality of the surge suppressors accommodated in the base unit, for a multi-phase power system.

20. The surge suppressor assembly as claimed in claim **19**, which defines a common earth connection and in which the plurality of surge suppressors are electrically connected to the base unit, thereby to be electrically connected to the common earth connection.

21. A surge suppressor kit for assembling into a surge suppressor assembly as claimed in claim **18**, in which a plurality of surge suppressors are provided with different ratings or characteristics, such that the surge suppressor assembly can be constituted using the base and one or more surge suppressors selected from plurality of surge suppressors provided with the different ratings or characteristics.

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